

PILLAR 01 // OPERATIONAL ADVANTAGE

HIGH-ALTITUDE FORWARD RESUPPLY & TACTICAL LOGISTICS

Runway-Independent Autonomous Heavy-Lift Airlift for Forward Himalayan Posts (LAC / LoC)


Ecosystem Partner: **BCN Drone Center (Spain)**

TACTICAL FIELD INTEGRATION TRIAL

FIGURE 1.0: FORWARD OPERATING BASE (FOB) RESUPPLY TRIAL — 4,800M AMSL



RAPID MODULAR CARGO POD INTEGRATION AT HIGH ALTITUDE
Field technicians and forward personnel conducting rapid pod exchange in sub-zero alpine conditions. The hermetically sealed bay accommodates emergency blood bags, tactical communications batteries, and precision resupply payloads.

<90 SEC
POD SWAP TIME

5-STAGE AUTONOMOUS LOGISTICS WORKFLOW

END-TO-END AUTOMATED MISSION CYCLE

STAGE 01
PRE-FLIGHT LOAD
Quick-lock mounting of insulated medical containers, ammunition canisters, or dual tactical batteries without specialized tooling.

STAGE 02
ZERO-RUNWAY VTOL
Four high-torque carbon rotors execute vertical liftoff from unprepared mountain clearings, transitioning to fixed-wing flight at 50 kts.

STAGE 03
ALPINE CRUISE
High-efficiency rear pusher motor cruises along pre-programmed mountain valley corridors at 110 km/h with active terrain avoidance.

STAGE 04
AUTONOMOUS DELIVERY
Precision hover descent or autonomous pod release at isolated forward posts with sub-meter GPS/NAVIC coordinates.

STAGE 05
SECURE BVLOS RTB
Automated return-to-base along alternate terrain-masked tracks with low thermal and acoustic signature throughout.

SPEC 01 • ENVIRONMENT
-20°C TO +50°C
HIGH-ALTITUDE CEILING
Engineered for density altitudes exceeding 5,500m AMSL. Carbon-structure heating prevents icing on leading edges and pitot tubes during severe Himalayan winter blizzard operations.
5,500M CEILING | ANTI-ICING | HIMALAYAN FIT

SPEC 02 • MOBILITY
ZERO FOOTPRINT
RUNWAY-FREE LAUNCH
Requires no catapults, runways, or recovery nets. Can deploy directly from forward outpost helipads, ridge trenches, or tactical 4x4 flatbed mobile support units.
ZERO RUNWAY | RIDGE DEPLOYMENT | 4X4 COMPATIBLE

SPEC 03 • PAYLOAD
MODULAR POD
HERMETIC CARGO BAY
Aerodynamic composite fuselage accommodates swappable cargo modules: active temperature-regulated medical pods, tactical supplies, or EO/IR surveillance gimbals.
RAPID SWAP | COLD-CHAIN FIT | TACTICAL PAYLOAD

SPEC 04 • AVIONICS
MANET MESH
SECURE BVLOS LINK
Frequency-hopping spread spectrum (FHSS) telemetry with AES-256 military-grade encryption. Seamless failover between NAVIC satellite navigation and optical inertial dead-reckoning.
AES-256 | NAVIC DUAL-BAND | ANTI-JAM MESH